Introduction to Data Science

What is Data Science?

Data Science is an interdisciplinary field that combines statistics, programming, and domain knowledge to extract meaningful insights from structured and unstructured data.

The Data Science Pipeline

The typical data science workflow includes data collection, data cleaning, exploratory data analysis, modeling, evaluation, and deployment.

Tools and Technologies

Python and R are the most popular programming languages used in data science projects.

Key Python libraries include Pandas for data manipulation, NumPy for numerical computing, and Matplotlib and Seaborn for data visualization.

Power BI and Tableau are popular tools for building interactive dashboards and business intelligence reports.

Big Data and Cloud

Apache Spark and Hadoop are used to process large scale datasets that cannot fit in memory.

Cloud platforms like AWS, Google Cloud, and Azure provide scalable infrastructure for deploying data science workflows.

Career in Data Science

Data scientists are in high demand across industries including healthcare, finance, retail, and technology.